Fiscal Year:	FY 2010	**Task Last Updated:**	FY 12/02/2009
PI Name:	Goodwin, Edwin Ph.D.		
Project Title:	Chromatid Painting for Chromosomal Inversion Detection		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	INDUSTRY	**Phone:**	(970) 493-066
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City:	Laporte	**State:**	CO
Zip Code:	80535-9750	**Congressional District:**	4
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	SBIR Phase II
Start Date:	01/22/2010	**End Date:**	03/21/2011
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Cucinott1a, Francis	**Contact Phone:**	281-483-0968
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Flight Program:			
Flight Assignment:	NOTE: Correct period of performance is 1/22/10-3/21/11 (previously listed as 12/1/09-11/30/11) and grant number change per J. Marsack/JSC (11/10)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX10CB05C		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	We propose the continued development of a novel approach to the detection of chromosomal inversions. Transmissible chromosome aberrations (translocations and inversions) have profound genetic effects, such as disrupting regulatory sequences that control gene expression, or creating genetic chimeras. These chromosome aberrations play a causative role in cancer, and ionizing radiation is one of the most efficient agents known to induce them. As such, chromosome aberrations are relevant to three NASA needs, biodosimetry, analysis of astronaut lymphocytes for cumulative radiation damage, and space radiation risk modeling. Of all structural chromosomal anomalies, inversions Â– a reversal of orientation of material within a chromosome Â– are the most difficult to detect. This is especially true of small inversions, most of which are invisible to all current cytogenetic techniques. Yet small inversions are likely the most transmissible (nonlethal) form of chromosomal damage, so they persist for long periods. This is a useful feature for		

	retrospective biodosimetry, and may also prove to be useful as an indicator of radiation quality.. In Phase 1 we demonstrated the use of a human chromosome 3, partial chromatid paint to detect a known inversion. During Phase 2, we will continue to improve the efficiency of the technology, an essential goal for commercialization (Phase 3) ultimately creating an improved and complete chromatid paint for chromosome 3. Finally, we will test the chromosome 3 'chromatid paint's' ability to detect radiation-induced inversions, and establish their frequency. The technology readiness level at the end of the Phase 2 contract is expected to be 5, i.e. validated in laboratory and relevant environments.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Three applications of interest to NASA are: 1) Retrospective biodosimetry, the estimation of radiation dose based on observed biological damage at a time point considerably after the exposure; 2) Ongoing chromosomal analysis of astronaut lymphocytes provides a measure of accumulated genetic damage caused by space radiation exposure. Chromatid paints will increase sensitivity of this analysis by adding a new class of observable aberrations; and 3) Biology-based space radiation risk analysis efforts. Chromosome aberrations play a causative role in carcinogenesis, as does gene copy number imbalance and cytogenetically invisible point mutations. One way to bring knowledge of cancer genetics into risk analysis is to estimate dose- and LET-dependent probabilities for specific cancer-related genetic alterations. Although some large inversions that can be detected are known to be associated with certain cancers, adequate investigation of cancer-specific chromosomal inversions is currently not possible, and therefore many such inversions, especially small ones, undoubtedly remain undiscovered. This shortcoming is pertinent with regard to charged particle exposures in that small inversions are likely to be among the most common and most stable chromosome aberration created. Chromatid paints have the potential to contribute to biology-based risk analysis through their ability to reveal these previously cryptic, currently predicted small cancer-related inversions for the first time. We expect that chromatid paints will fulfill multiple needs in several fields related to human health. A major market for inversion detection is the screening of cancer cells for previously undetected inversions that are causal. New cancer specific inversions may lead to diagnostic and prognostic tests and may even lead to eventual drug targeting. Other markets include, but are not limited to: clinical cytogenetics (cancer diagnosis and prognosis, detection and diagnosis of certain neurological disorders associated with inverted copy number variants, infertility diagnosis and genetic counseling); biomedical research (mechanistic studies of cancer induction, radiation effects, and chemical toxicology); and governmental agencies (retrospective biodosimetry-evaluation of radiation exposure as might occur accidentally or from a terrorist attack). Ultimately, multi-color chromatid paints will provide genome-wide translocation identification, just as mFISH and SKY currently do, and in addition will allow simultaneous and sensitive detection of inversions, making it possible, even probable, that chromatid paints may eventually capture much of the market currently held by chromosome paints.
Task Progress:	New project for FY2010. Reporting not required for this SBIR Phase 2 project.
Bibliography Type:	Description: (Last Updated: 08/14/2015)
Articles in Peer-reviewed Journals	Ray FA, Robinson E, McKenna M, Hada M, George K, Cucinotta F, Goodwin EH, Bedford JS, Bailey SM, Cornforth MN. "Directional genomic hybridization: inversions as a potential biodosimeter for retrospective radiation exposure." Radiat Environ Biophys. 2014 May;53(2):255-63. Epub 2014 Jan 30. http://dx.doi.org/10.1007/s00411-014-0513-1 ; PubMed PMID: 24477407 , May-2014